Forecasting China's CPI with Mixed-Frequency Data: A Comparative Study of Machine Learning Approaches

ECO6130 Final Project – Group 2

Yiran Liu, Xinyi Han, Xinyi Zhuo, Yingzhu Song, Mansha Cai

April 19, 2026

Outline

Outline

1.1 Background: The Data Frequency Dilemma in Macro Forecasting

Traditional Macro Forecasting:

- **The Crucial Role of CPI:** Serves as the primary anchor for PBOC's monetary policy formulation, asset pricing, and macroeconomic risk assessment.
- **The "Low-Frequency" Trap:** Traditional forecasting models rely heavily on monthly or quarterly aggregated variables (e.g., PPI, social financing). They inherently fail to capture rapid, high-frequency microeconomic dynamics within a month.

The Core Pain Point: Publication Lag & Data Asynchrony

- **Real-world reporting delays:** Macro indicators are published with significant lags (e.g., China's NBS merging Jan/Feb data due to the Spring Festival).
- **The "Ragged Edge" Problem:** In real-time nowcasting, synchronous data is often unavailable, rendering traditional models practically useless for immediate decision-making.

1.2 Research Objectives & Methodological Framework

Core Research Objectives:

1. To construct a robust mixed-frequency pipeline that strictly prevents "look-ahead bias" caused by data publication lags.
2. To evaluate the efficacy of non-linear machine learning algorithms under the constraints of small-sample macroeconomic datasets.

Our Workflow Pipeline:

Raw Mixed-Frequency Data → 15-Month Offset Window (Anti-Leakage) → Feature Engineering → Time-Series Cross Validation → Economic Attribution

Multi-Model Evaluation Matrix:

- **Linear Baseline (PCA + OLS):** To test whether linear assumptions hold.
- **Ensemble Learning (Random Forest):** To mitigate small-sample overfitting via Bagging.
- **Deep Learning (Dual-Encoder LSTM):** To capture complex cross-frequency patterns.

Outline

2.1 Data Sourcing & The Mixed-Frequency Matrix

Sample Horizon & Source:

- **Period:** March 2017 – March 2025 (Yielding ∼81 effective monthly samples).
- **Source:** All data acquired via the *iFind* financial database.

High-Frequency Indicators (Daily/Weekly)

- **Goal:** Capture rapid micro-shocks.
- **Examples:** Pork Wholesale Prices (Pig cycle), Subway Ridership (Offline activity), OMO Net Injection (Liquidity).

Low-Frequency Indicators (Monthly)

- **Goal:** Anchor macro fundamentals.
- **Examples:** Social Financing, Export Growth, PPI (Consumer/Producer Goods).

2.2 Data Preprocessing & Look-Ahead Bias Prevention

Step 1: Cleaning & Temporal Scaling

- **Logical Imputation:** Handled missing OMO data as zero-imputation.
- **Target Alignment:** Excluded Jan/Feb merged CPI; used Jan-Mar high-frequency average to predict March CPI.
- **Ten-day Slicing:** Aggregated daily data into early, middle, and late 10-day periods.

Step 2: The 15-Month Offset Sliding Window (Crucial Anti-Leakage)

- **The Reality:** Macro data for Month T is unpublished when forecasting Month T.
- **Our Mechanism:** To predict CPI in Month T, the model utilizes High-Frequency data up to Month T, but **strictly limits Low-Frequency data to Month $T - 1$**.
- **Impact:** Completely **eliminates look-ahead bias**, perfectly simulating real-world nowcasting.

2.3 Exploratory Data Analysis: Correlation Insights

- **Price Transmission:** A strong correlation ($r = 0.78$) between Consumer PPI and CPI validates the industrial upstream-to-downstream mechanism.
- **Multicollinearity:** Significant correlations among high-frequency variables (e.g., cement, housing) justify the use of PCA for dimensionality reduction.
- **Non-Linearity Rationale:** The weak linear relationship between high-frequency signals and CPI necessitates non-linear machine learning architectures.

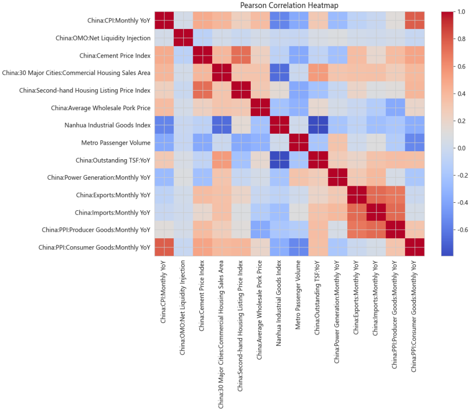

Figure 1. Pearson Correlation Heatmap of CPI and Mixed-Frequency Features

Outline

3.1 Model Selection Matrix & Research Objectives

Table 1. The Model Evaluation Matrix

Model	Why Chosen	Main Idea
PCA + OLS	Linear Benchmark	Applies OLS to extracted macro factors for a transparent baseline.
Random Forest	Small-Sample Robustness	Uses Bagging and feature flattening to capture non-linearities and mitigate noise.
Dual-Encoder	Mixed-Frequency Design	Employs a bifurcated architecture for parallel representation and cross-frequency fusion.

Research Objectives & Logic:

- **Testing Non-linearity:** To verify whether China's CPI evolution is driven by simple linear patterns or complex non-linear dynamics.
- **Balancing Complexity:** To find the optimal equilibrium between architectural depth and small-sample constraints (81 months).
- **Strategic Goal:** To identify the most robust framework for translating mixed-frequency economic noise into actionable forecasts.

3.2 Linear Baseline Construction: PCA + OLS

Step 1: Unsupervised Factor Extraction & Standardization

- **Principal Component Analysis (PCA):** Utilized to filter high-frequency noise and extract orthogonal "latent macro factors".
- **Isolated Standardization:** Implements a strict feature pipeline where scaling parameters are fitted exclusively on the training set to prevent any "look-ahead bias".

Step 2: The Linear Benchmark

- **Linear Mapping:** Applies **Ordinary Least Squares (OLS)** to the reduced 210-dimensional macroeconomic feature vector to establish a transparent linear baseline for performance comparison.

3.3 Non-linear Ensemble Construction: Random Forest

1. Bagging Paradigm

Constructs multiple independent decision trees to aggregate predictions, effectively reducing variance on high-dimensional datasets.

---

2. Small-Sample Regularization (Crucial constraints)

Strictly limits tree depth (`max_depth=5`) and leaf node samples (`min_samples_leaf=3`) to mitigate the risk of fitting macroeconomic noise.

---

3. High-Dimensional Feature Flattening

Processes the 15-month sliding window into a 210-dimensional cross-sectional space (15×14), leveraging the model's natural scale-invariance to skip standardization.

3.4 Deep Learning Exploration: Dual-Encoder Network

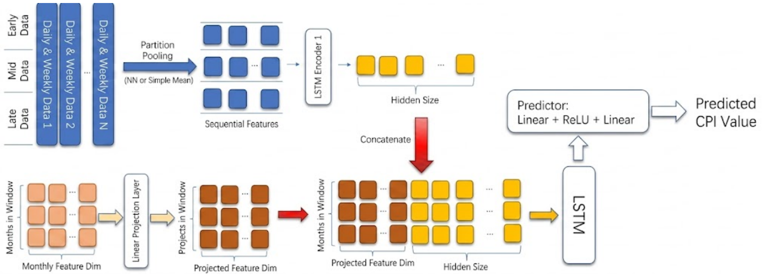

Figure 2. Architecture of the Dual-Encoder Feature Fusion Network

Bifurcated Architecture Design:

- **HF Encoder:** Utilizes an LSTM network to process 10-day sequences, acutely capturing micro-transient shocks.
- **LF Encoder:** Employs a Linear Projection Layer to track stable macro fundamentals.

Fusion & Robustness:

- **Cross-Frequency Fusion:** A Temporal Fusion Layer concatenates latent branches to extract complex non-linear interactions.
- **Regularization:** Incorporates Huber Loss (outlier resistance) and Dropout (0.2) to alleviate small-sample constraints.

Outline

4.1 Multi-Model Performance Evaluation & Discussion

Table 2. Average Out-of-Sample Performance Comparison (3-fold Time-Series CV)

Model Category	**Specific Model**	**MSE**	**MAE**	**RMSE**	R^2
Ensemble Learning	**Random Forest**	**0.6545**	**0.5407**	**0.6508**	**0.1872**
Deep Learning	Dual-Encoder (LSTM)	0.9525	0.7527	0.8805	-0.7337
Linear Model	PCA + OLS	2.5931	1.3997	1.6042	-9.9784

1. **Ensemble Superiority:** Random Forest achieves the highest precision and is the only model yielding positive explanatory power.
2. **Linear Paradigm Failure:** Extreme negative R^2 indicates linear mappings cannot characterize complex nonlinear CPI dynamics.
3. **DNN Bottlenecks:** Dual-Encoder suffered from overfitting due to "parameter-data asymmetry" in a small 81-month sample.

4.2 Visualizing Forecast Trajectories & Inflection Points

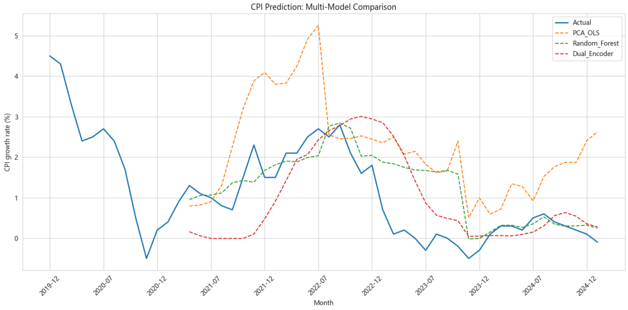

Figure 3. Time-Series Comparison of Multi-Model Fitting

- **RF Synchronicity:** Maintains high temporal alignment with realized CPI, anchoring medium-term volatility.
- **DL Phase Lag:** Exhibits "over-smoothing" and a 1-2 month lag at critical price inflection points.
- **Linear Deviation:** PCA+OLS deviates severely, validating the failure of linear mapping in turning points.

4.3 Feature Attribution & Economic Logic Validation

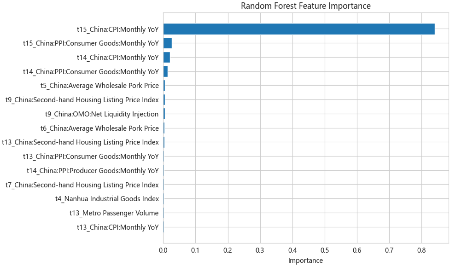

Figure 4. Random Forest Feature Importance Bar Chart

1. **Inflation Inertia:** Lagged CPI terms account for 86.3% weight, corroborating strong path dependence.
2. **Vertical Transmission:** Consumer PPI validation of industrial chain transmission to retail CPI.
3. **Pig Cycle:** Pork prices at 10-11 months lag perfectly align with biological breeding cycles.
4. **High-Freq Value:** Subway ridership captures short-term micro-shocks as signal confirmation.

4.4 Analysis of Seasonal Volatility & Adaptive Potential

Table 3. Quarterly Error Statistics

Quarter	Error (%)	Std (%)	Samples
Q1	+0.477	1.185	4
Q2	+0.149	1.191	11
Q3	-0.086	0.621	12
Q4	+0.093	1.094	12

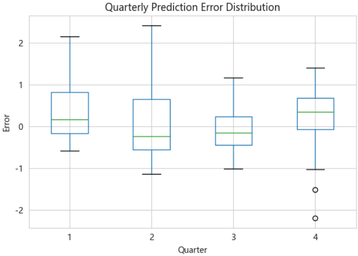

Figure 5. Quarterly Error Distribution

- **Q1 Challenge:** Pronounced positive bias (+0.477%) due to consumption volatility and data sparsity during the Spring Festival period.
- **Q3 Robustness:** Highest predictive precision (Std Dev = 0.621%) achieved during stable macro environments with minimal seasonal shocks.
- **Adaptive Potential:** Dual-Encoder's RMSE improved to 0.306 in 2022-2023, revealing neural networks' capacity to adapt to structural economic shifts.

Outline

5.1 Empirical Interpretation & Core Conclusions

Empirical Interpretation

- **Inflation Inertia:** Lagged Consumer Price Index terms are primary drivers, reflecting strong path-dependence.
- **Price Transmission:** Producer Price Index ranking validates the pass-through mechanism to retail markets.
- **Biological Regularity:** Pork prices at a 10-11 month lag accurately map the biological breeding cycle.
- **Micro-data Utility:** High-frequency indicators provide signal confirmation but remain secondary to macro anchors.

Research Conclusions

- **Optimal Architecture:** Random Forest is the superior choice for small-sample macro forecasting and generalization.
- **Mixed-Frequency Value:** Integrating high-frequency data significantly enhances timeliness and provides incremental value.
- **Model Bottleneck:** Deep learning potential is restricted by the limited 81-month macroeconomic sample size.

5.2 Reflections on Limitations & Future Research

Model Limitations

- **Data Capacity:** Small macro samples and model complexity lead to overfitting risks.
- **Feature Gap:** Numerical data fails to capture sudden policy shifts or geopolitical shocks.
- **Static Fusion:** Fixed fusion logic cannot adapt feature weights across varying economic cycles.

Future Directions

- **Multimodal Expansion:** Integrate Natural Language Processing to extract policy signals from central bank reports.
- **Transfer Learning:** Use international datasets for pre-training to alleviate small-sample pressures in China.
- **Attention Mechanisms:** Introduce Attention layers to schedule feature weights under specific economic backgrounds.

Thank You!

ECO6130 Final Project – Group 2